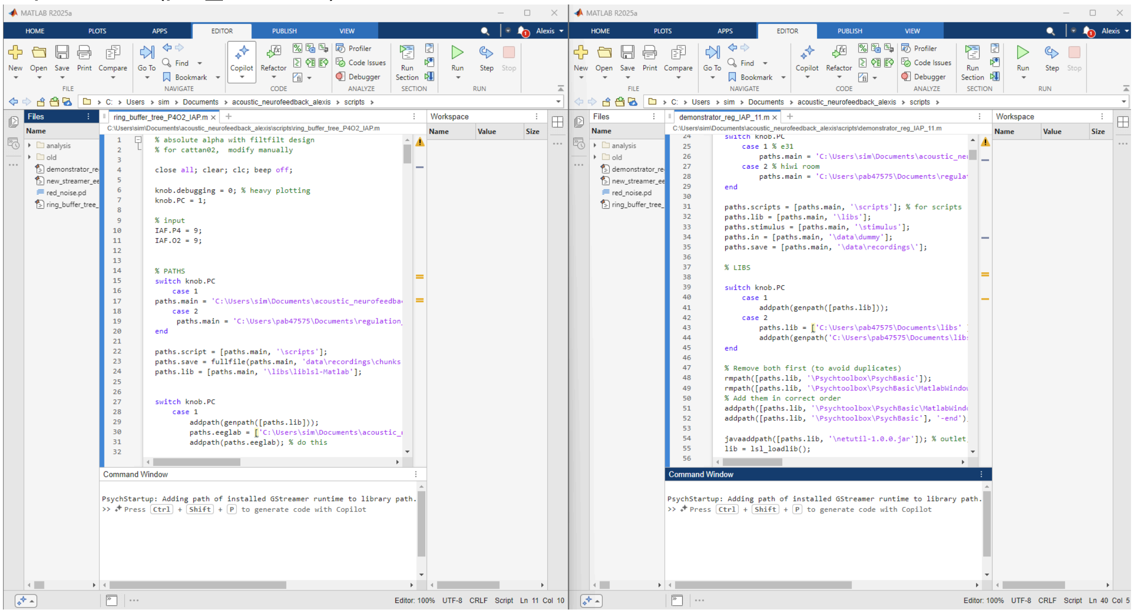
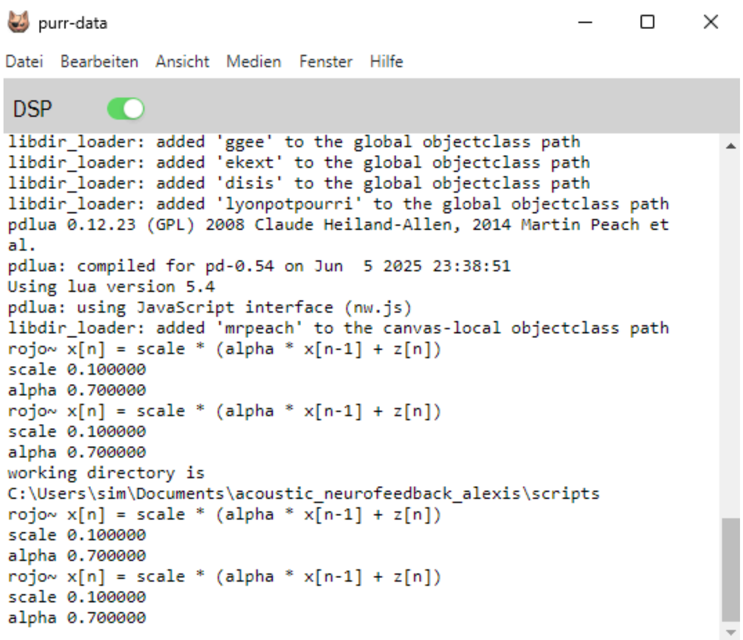
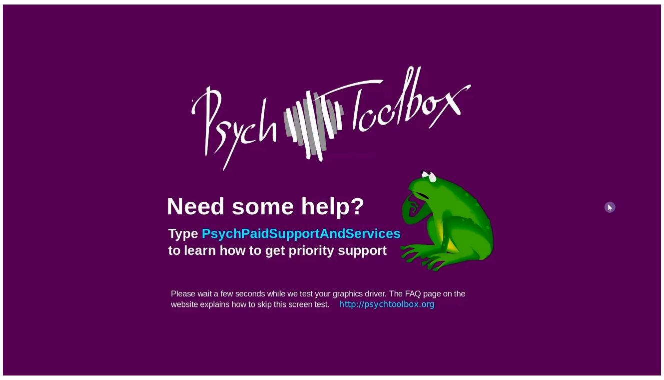
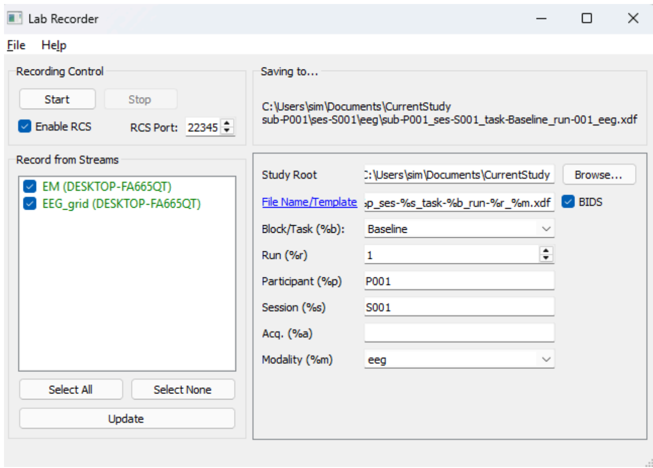
<table>
<tr><th colspan="2">Presenter PC:</th></tr>
<tr><td>1.1</td><td>
1. Extend view (it’s usually duplicated): Windows+P → Erweitern<br>
2. Test keyboard inside the room<br>
3. Sound system:<br>
    a. Switch on speakers and attenuator<br>
    b. Speakers’ wheel on “LINE”<br>
    c. PC volume: 40% & Lautsprecher<br>
    d. Speakers volume: white stripes<br>
4. Folder with all scripts:<br>
\\smb.uni-oldenburg.de\psychologie$\Neuro\data\projects\alo_alpha_neurofeedback\code\experiment
</td></tr>
<tr><td>1.2</td><td>
MATLAB scripts: open in 2 <u>different</u> MATLAB <u>instances</u> (no tabs in one instance) (as in the picture below)<br>
- Online Analysis (ring_buffer.m)<br>
- Experiment (ptb_demo.m)<br>

</td></tr>
<tr><td>1.3</td><td>
Open Purr Data patch: red_noise.pd<br>
There will be 2 windows, <u>minimize</u> the bigger one and leave the smaller (picture → )<br>
<br>
To activate the audio there are 2 options<br>
1. DSP green toggle<br>
2. → Medien → An<br>
<br>
If you want to double-check if the audio works:<br>
→ Medien → Teste Audio → OUTPUT MONITOR options<br>
<br>
If the audio sounds awful, a possible solution is:<br>
→ Einstellugen → Ausgabe-Geräte → Lautsprecher (Sound Blaster X)<br>


</td></tr>
</table>

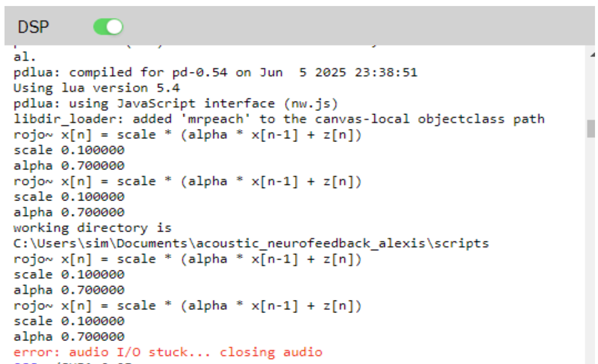
<table>
<tr><th colspan="2">Start</th></tr>
<tr><td>2.1</td><td>Prepare participant & EEG cap<br>
- Open mbt, connect amplifier and stream signal at 250 Hz</td></tr>
<tr><td></td><td>2 options to run simulations without participant<br>
1. Open mbt and run the dummy signal<br>
2. Stream an existing dataset:<br>
- Open a third matlab instance with new_streamer.m (same folder as ring_buffer.m and ptb_demo.m)<br>
- Choose dataset to stream (line 7) and press RUN<br>
- Wait, you’ll see a plot (showing O2 channel of participant’s EEG) and the Command WIndow will let you know when it’s running.</td></tr>
<tr><td>2.2</td><td>Run Online Analysis (ring_buffer.m)<br>
- Command Window will tell you when it’s ready to start the experiment control script</td></tr>
<tr><td>2.3</td><td>Run Experiment Control Script (ptb_demo.m)<br>
- A purple screen with a green frog should appear on participant’s screen, followed up by a Welcome message asking participants to wait.




- Start the LabRecorder App once the MATLAB command window tells you.<br>
  - Select all streams: EEG, AAA, EM, NFB<br>
- Inform participants that they can start by pressing the [spacebar]<br>
- During the experiment you don’t need to do anything: the command window will keep you updated of participant’s progress through the instructions and trials.


</td></tr>
<tr><td>2.4</td><td>Purr Data patch troubleshooting:<br>
If “error: audio I/O stuck… closing audio”<br>
- close and open purr-data<br>
- this won’t affect the recording, you don’t need to restart.

Click out of the window before starting so when the participant press the [spacebar] they won’t toggle the DSP accidentally.


</td></tr>
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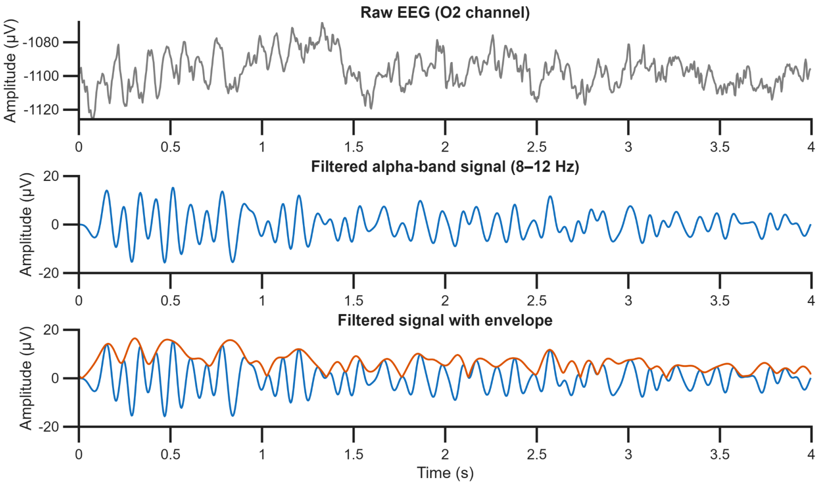
<table>
<tr><th colspan="2">End</th></tr>
<tr><td>3.1</td><td>

** Important: the following order is relevant to ensure a proper offline reconstruction*

**1.** Stop streaming signals in mbt (smarting streamer app)

**2.** The online analysis script (ring_buffer.m) should end **automatically**

- The command window will tell you where the chunksize got saved (line 13)
- The plot below should pop-up
- If not, you need to run the last section manually (line 280)




**3.** Stop LabRecorder

</td></tr>
<tr><td>3.2</td><td>

The experiment control script (ptb_demo.m) Command Window will show where the variables got saved

- Variables: score.mat, trial.mat, task.mat, thresh.mat and storage.mat
- If it didn’t run till the end, you need to run the last section to save those variables (line 2350).
- Move out these variables fast, they will get rewritten if you do another recording.

You need to manually close the experiment control script (ptb_demo.m).

- Close the last screen with [Esc].

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